How Economists Discuss Central Bank Digital Currencies

Themes, Tone and Stance, 2014–2026

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M2 – Private crypto
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Research question

"How has economists and institutional discourse on CBDCs evolved between 2014 and 2026, and to what extent does this evolution align with changes in the policy environment?"

Abstract

Keywords:
Number of pages: 9.3 (main text, Introduction to Conclusion, figures excluded)
Appendix: 10 pages

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1 Introduction

2 Related literature

2.1 Central bank digital currencies

2.2 What the field builds on

2.3 Bibliometric and text-as-data studies of CBDCs

Within our corpus, we observe 29 titles that announce a literature, systematic or bibliometric review of CBDCs, published between 2019 and 2026. When comparing the five most recent bibliometric studies with ours, we found three shared main limits. First, their data come from one or two databases, Scopus and/or Web of Science, and are mostly journal papers written in English. Working papers, central-bank speeches and central-bank documents are therefore mostly absent from their analysis, so that their sample is very limited compared to ours: a few hundred documents compared to 4,710. In addition, they build their analysis with standard tools that mainly describe what is studied, but not how (tone and stance) or why. Finally, their classifications are never checked against a reading of the texts themselves.

Our study addresses these three limits. Our corpus is much larger, with papers coming from five databases, central-bank speeches and documents, in different languages, and every text is checked to be written by an economist or an economic institution. We ask not only what is written but also who writes it, how, why and when. Finally, each automatic classification we use is compared with our own reading of a sample of texts.

MISSING : text-as-data studies

3 Data

3.1 Sources and collection

In order to properly use the tools and for better selection purposes, we try first to fetch as many documents as possible. Indeed, if we had selected the sample based on pre-determined criteria of our choice, we could face the bias of pre-selecting the sample. By having a larger sample with partly irrelevant data, we can apply selection processes based on quality criteria, such as the relevance of a document. Also, it is useful to have a large sample to be able to validate results on sub-samples or parts of the data kept for testing, which strengthen the results, or for robustness tests.

When building the fetching algorithm, we therefore tried to reach for selection criteria as broad as possible. First, we wanted many "types" of documents, namely: official academic papers published in journals or magazines, press articles, speeches from institutional figures or official documents. The sources span :

- Academic databases: OpenAlex, Semantic Scholar, arXiv, HAL and DOAJ, which mainly contain working papers and journal articles, but also books.
- Institutional databases: speeches were mainly extracted from the BIS archive (as well as BIS RSS) and ECB's speech database. We also extracted from 26 central banks', the EU institutions and international institutions sites the pages relating to CBDC.
- Press: press articles are extracted from Google News

The searching process is fairly easy: we choose 115 sentences in 12 langues (using LLM and suggestions, as well as our own formulations, based on common expressions in the literature), that are

used to search accross titles and abstracts alike. The APIs return results page by page for the papers and authorities' websites and the press articles are harvested using a recursive bisection to single down the relevant articles. Everything from all sources is collected and we can go over the filtering process.

3.2 De-duplication and enrichment

Because some articles may be duplicates or pre-publications or have similar names, we try to get rid of all duplicates or fuse when they are from the same authors and relate to the same paper. The contents of the documents are also enriched, in that we also harvest the language in which it is procuded, the keywords of the text (using YAKE), the exact date (when possible, if not indexed at the beginning of the year issued), the authors' names and media in which it has been issued, the introduction and conclusion for the papers, which resume the motivations and results of the paper, as well as the content of the text.

3.3 Relevance: is CBDC the theme?

Because CBDC is also a skin disease (Chronic Bullous Disease of Childhood) and a chemical term (CycloButane DiCarboxilate), or that it can be used as an acronym for anything that the author does not explicit in the titles and abstracts, it is possible to obtain documents unrelated to the monetary concept. Hence, using a range of contextual vocabulary (such as currency, payment, central bank, monnaie, Geld, etc.), we filter for papers in which the term CBDC is ambiguous and not directly related to Central Bank Digital Currency.

More importantly, we want CBDC to be the main theme of the paper, and not merely a cited reference related to the actual content. Therefore, we establish relevance scoring as such: each field $f \in F$ gets centrality score $c_f \in [0, 1]$ (this term will be also used later in the paper, but with a different meaning, be cautious !) that gives weight to the paper relevance if CBDC is a central theme to the paper. The centrality scores are determined as such:

$$c_{title} = \mathbf{1}[\text{CBDC in title}] \qquad c_{kw} = 0.4 \cdot \mathbf{1}[\text{CBDC keyword}]$$
$$c_{abs} = \frac{\text{card(sentences mentioning CBDC)}}{\text{card(sentences)}} \qquad c_{body} = 0.5 \min\left(1, \frac{\rho}{3}\right) + 0.5\sigma$$

Where ρ = mentions per 1000 words and σ = share of ten slices with a mention. And the relevance score is:

$$\text{score} = 100 \cdot \left(0.6 \max_{f \in F} c_f + \frac{0.4}{|F|} \sum_{f \in F} c_f \right) \tag{1}$$

3.4 Who counts as an economist

We want now to ensure that the text has been written by an economist. This is fairly easy since we can just check using the authors' ID for each academic sources, which tells whether the author is an economist or not. Therefore, all research document have been kept, when the relevance score clearly indicated that the theme was CBDC. Moreover, since articles and speeches issued by authorities are produced by either economists working there or economists by practice (C. Lagarde is a lawyer for example, but her vast experience as head of IMF and of ECB qualifies her as such). Press articles and documents produces by governments were therefore dropped.

3.5 Descriptive statistics

The corpus analysed in this paper holds the 4,710 texts with CBDC as a theme that were written by economists (Table 1). Research makes up 85% of it (3,994 texts); central bank speeches (511) and documents of central banks and international institutions (205) make up the rest. For three texts in four, CBDC is the main theme.

The first texts by economists date from 2016, and the field took off after 2019 (Figure 1). Research grew from 20 texts in 2017 to 839 in 2025, a compound rate of about 60% a year, and 2026 had already exceeded 2025 by October. Speeches rose until 2022 (77) and have stayed close to 70 a year since. Documents of institutions jumped in 2023 (49), the year of the European Commission's proposal for a digital euro and of the end of the ECB's investigation phase. The first period of the paper, 2014–2019 rests on a small number of texts (211 texts, against 972 in 2020–2022 and 3,360 in 2023–2026), which is why changes are estimated with intervals and tested against random dates (Sections 6 and 8).

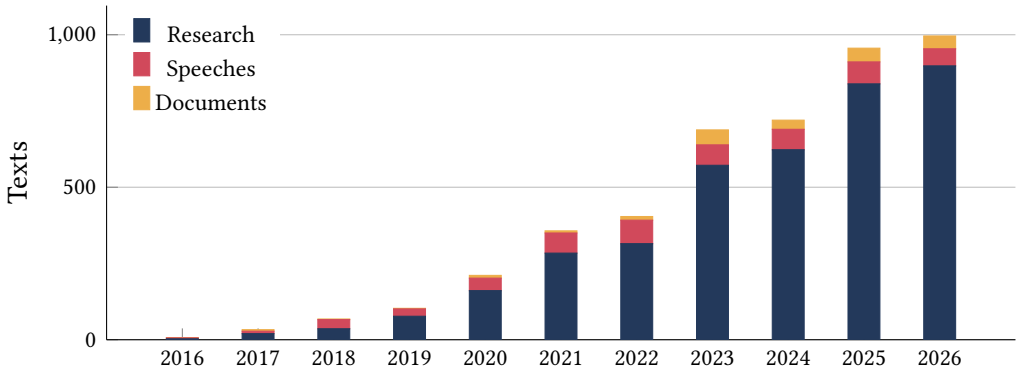


Figure 1. Texts written by economists, by year and type (2026: until October)

Research is spread over 1,479 outlets, and working papers weigh as much as journals: the largest outlets are SSRN (461 texts) and RePEc (164), ahead of any journal. Articles account for 2,455 research texts, preprints for 583, book chapters for 394. Citations are highly concentrated: the median research text has not yet been cited, while the mean is 5.5 citations, so a small core of papers receives most of the attention (Section 4). On the institutional side, 61 institutions are represented, led by the European Central Bank (216 texts), the Deutsche Bundesbank (84), the Banque de France (49), the Bank of Japan (41) and the Sveriges Riksbank (32): the digital euro makes the euro area the centre of the institutional debate.

Texts are written in 39 languages. English dominates (79%), but Russian is a substantial second language (422 texts, 9%), mostly on the digital ruble, ahead of German, Ukrainian and Spanish. English-only databases miss about a fifth of the economists' writing on CBDCs.

Most texts can be dated to the day (71%), and most have keywords (85%). An abstract is available for 68% of texts, an introduction for 62% and the full text for 63%, while reference lists, needed for co-citation, exist for 37% of research texts only. Each analysis uses the texts for which its input exists, and the number of texts is reported with every result.

Table 1. The economists' corpus

	Texts		Texts
Type of text		*Language (39 in all)*	
Research	3,994	English	3,721
Central bank speeches	511	Russian	422
Documents of institutions	205	German	94
Total	4,710	Ukrainian	91
Relevance		Spanish	74
CBDC the main theme	3,513	French	43
CBDC a secondary theme	1,197	*Fields available*	
Research, by document type		Exact day of publication	71%
Article	2,455	Abstract	68%
Preprint	583	Introduction	62%
Book chapter	394	Full text	63%
Conference paper	131	Keywords	85%
Dissertation	117	Reference list (research)	37%

4 Who writes

4.1 Method: co-citation, co-authorship, bibliometric laws and distinctive vocabulary

In the following analysis, to see who writes about CBDCs (which authors, institutions and papers), we use four different tools. The first one, called a co-citation network, links references that are cited in the same paper, which allows us to build groups of references often cited together. Then, we build a co-authorship network, which shows research teams by linking authors when they write a paper together. In both networks, to avoid giving too much weight to links between very popular references or authors, we weigh each link by its association strength, which takes into account how popular each item is.

For two items i and j (references or authors), let c_{ij} be the number of papers in which they appear together and c_i the number of papers in which i appears. The weight of the link is the association strength

$$s_{ij} = \frac{c_{ij}}{c_i\, c_j}, \tag{2}$$

proportional to the ratio of observed to expected co-occurrences (van Eck and Waltman, 2009). Groups are found with the Louvain algorithm, which maximises the modularity

$$Q = \frac{1}{2m} \sum_{i,j} \left[s_{ij} - \frac{k_i k_j}{2m} \right] \delta(g_i, g_j), \tag{3}$$

where $k_i = \sum_j s_{ij}$, $m = \frac{1}{2} \sum_{i,j} s_{ij}$, and $\delta(g_i, g_j) = 1$ when i and j are in the same group.

Besides these two networks, we also compute three standard bibliometric laws. To see how fast literature on CBDC grows, we compute the growth rate of published papers. The second one, Lotka's law, allows us to see how papers are spread among authors, whether a few authors write most of the papers or whether many authors only write one. Then, the last one, called Bradford's law, ranks publication sources by their number of papers on CBDCs and splits them into three groups of equal sizes in papers. This shows whether the literature is concentrated in a few sources or spread over many of them.

With N_y papers in year y, the compound growth rate between y_0 and y_1 is

$$g = \left(\frac{N_{y_1}}{N_{y_0}}\right)^{1/(y_1 - y_0)} - 1. \tag{4}$$

Lotka's law states that the share of authors with n papers decreases as a power of n,

$$f(n) = \frac{C}{n^{\beta}}, \tag{5}$$

with β estimated by least squares on $\log f(n) = \log C - \beta \log n$; $\beta \approx 2$ in mature fields, and a larger β means that more authors write a single paper. For Bradford's law, sources are sorted by decreasing number of papers and cut into three zones holding a third of the papers each; the number of sources in each zone, $n_1 : n_2 : n_3$, measures concentration.

Finally, in order to compare the words used by research economists and central bankers, we use a method called log-odds ratio with an informative Dirichlet prior. Compared with simple word counts, this method does not give too much weight to rare words and allows us to see which words one group uses more than the other.

Let y_{gw} be the number of times group g uses word w, n_g the total number of words of group g, and $\alpha_w = \alpha_0 \, \pi_w$ a prior proportional to the frequency π_w of w in the whole corpus. For group i against group j (Monroe, Colaresi and Quinn, 2008),

$$\delta_w = \log \frac{y_{iw} + \alpha_w}{n_i + \alpha_0 - y_{iw} - \alpha_w} - \log \frac{y_{jw} + \alpha_w}{n_j + \alpha_0 - y_{jw} - \alpha_w}, \qquad z_w = \frac{\delta_w}{\sqrt{\dfrac{1}{y_{iw} + \alpha_w} + \dfrac{1}{y_{jw} + \alpha_w}}}, \tag{6}$$

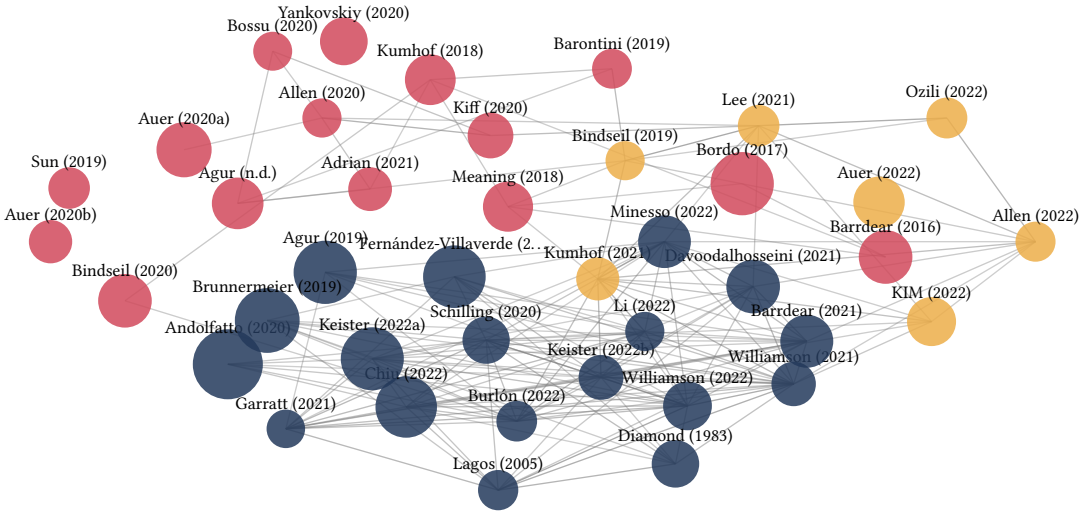
and words are ranked by z_w. The prior pulls the log-odds of rare words towards zero, so that a word used twice by one group and never by the other does not come first.

4.2 Findings

From our analysis, academic output on CBDCs grew by 60% a year between 2017 and 2025. The two largest sources are SSRN (461 papers) and RePEc (164 papers), two repositories of working papers. We also obtain a Lotka exponent of 3.3, which is higher than the value, of about 2, found in more mature fields: 83% of authors write only one paper on CBDCs. In the same way, Bradford's law shows that only 21 sources publish a third of the papers, while the last third is spread over more than 1,100 sources. In short, a small group of authors and sources publishes regularly on CBDCs, while most economists only write on the subject once.

Papers with at least one author from a central bank, the BIS, the IMF or the World Bank represent 10% of the papers but receive 48% of the citations, and their share of the papers fell from 23% in 2014-2019 to 7% in 2023-2026. In other words, these economists write few papers but are cited almost five times more than their weight in the literature, while most new papers now come from economists outside these institutions. The co-citation network gives three groups of references, around Andolfatto (2020) and Brunnermeier (2019), Bordo (2017) and Barrdear (2016), and Auer (2022). However, these groups are not clearly separated since we have a low of $Q = 0.10$: the literature does not split into distinct schools of thought. In addition, the co-authorship network shows that research teams are small: about half of the papers are written by a single author, and only 15% involve authors from different countries. Among institutions, the European Central Bank is by far the most active, with 118 speeches and 98 official documents. On the research side, Russia is the most represented country: it publishes the most

papers, and four of the eight most active research institutions are Russian.



Groups (largest members): • Andolfatto (2020), Brunnermeier (2019), Agur (2019), Keister (2022); • Bordo (2017), Auer (2020a), Barrdear (2016), Bindseil (2020); • Auer (2022), Kim (2022), Kumhof (2021), Lee (2021). 1,493 papers with reference lists, 13,810 distinct references; the 40 most co-cited references are shown; links weighted by association strength; Louvain clusters, $Q = 0.10$.

Figure 2. Co-citation network of research by economists

Finally, we observe that research economists and central bankers do not use the same words. Research economists use more words such as *legal*, *analysis*, *model* and *impact*, while central bankers use more words such as *payments*, *euro* and *Europe*. In other words, research economists seem to study CBDCs from a more analytical point of view, while central bankers mostly talk about payments and the digital euro.

Table 2. Words most over-used by each group

Research economists		Central bankers	
legal	23	payments	28
analysis	21	euro	22
development	21	european	18
research	20	europe	18
monetary	20	area	18
impact	19	services	15
model	18	innovation	14
implementation	17	conference	14

z-scores of the log-odds ratio with an informative Dirichlet prior, each group against the other. English texts only; research economists: title and abstract (259,105 words); central bankers: speeches (232,446 words). Words defining the topic (CBDC, digital, currency…) are removed.

5 What is written

5.1 Method: thematic map, topic model and covariate-adjusted shares

To see what is written by economists, we use two methods. The first one consists in grouping keywords that often appear together in the same papers into themes. Then, we place these themes on a thematic map with two axes: how much a theme is linked to the other themes (*centrality*) and how much its keywords are linked to each other (*density*). This allows us to classify themes into four types: motor, basic, niche, and emerging or declining.

Formally, with c_{ij} the number of papers where keywords i and j appear together and c_i the number of papers with keyword i, the equivalence index is $e_{ij} = c_{ij}^2/(c_i c_j)$. For a theme C (a group of keywords found as in Equation (3)),

$$\text{centrality}(C) = 10 \sum_{i \in C,\ j \notin C} e_{ij}, \qquad \text{density}(C) = \frac{100}{|C|} \sum_{i,j \in C} e_{ij}, \tag{7}$$

and the four quadrants are separated by the medians of the two measures (Callon, Courtial and Laville, 1991; Cobo et al., 2011).

The second method consists in grouping texts by their meaning. To do so, we use *SBERT*, a multilingual model that turns each title and abstract into a vector, so that two texts saying the same thing end up close, even if they are not written in the same language. We then group these vectors into 24 topics with *BERTopic* and label each topic with its most specific words.

Each text t is represented by a vector $\mathbf{e}_t \in \mathbb{R}^{384}$, and the similarity of two texts is the cosine

$$\cos(\mathbf{e}_s, \mathbf{e}_t) = \frac{\mathbf{e}_s \cdot \mathbf{e}_t}{\|\mathbf{e}_s\| \, \|\mathbf{e}_t\|}. \tag{8}$$

The vectors are reduced to five dimensions with UMAP and grouped with HDBSCAN, which finds dense regions without fixing the number of topics in advance. Each topic c is then described by the words with the highest class-based TF-IDF,

$$W_{w,c} = \frac{\text{tf}_{w,c}}{|c|} \, \log\left(1 + \frac{A}{f_w}\right), \tag{9}$$

where $\text{tf}_{w,c}$ is the number of occurrences of w in topic c, $|c|$ the number of words in c, f_w the number of occurrences of w in all topics and A the average number of words per topic (Grootendorst, 2022).

Finally, to make sure that a change in the share of a topic is not only due to a change in who writes, we run a logit regression of each topic on the period, the type of author and the language. That allows us to measure how the share of each topic changed between two periods (here, between 2014-2019 and 2023-2026), in percentage points, all else being equal.

For text t and an outcome $y_t \in \{0, 1\}$ (here, belonging to a topic),

$$P(y_t = 1) = \Lambda\big(a + b_{\text{period}(t)} + c_{\text{actor}(t)} + e \cdot \mathbf{1}[\text{English}_t]\big), \qquad \Lambda(x) = \frac{1}{1 + e^{-x}}, \tag{10}$$

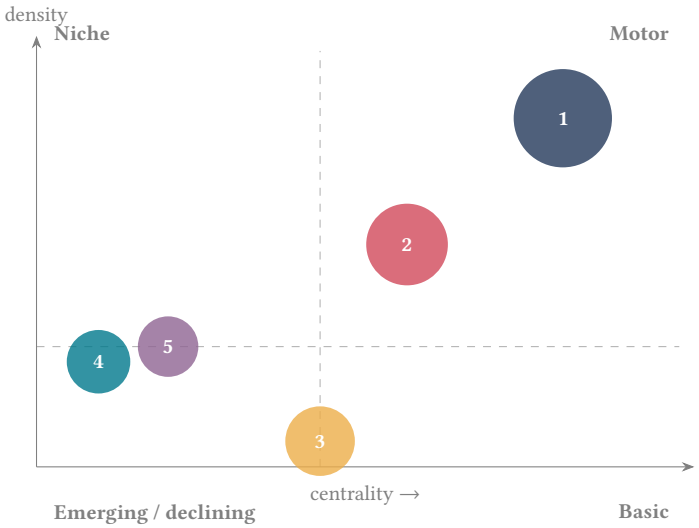
estimated with robust (HC1) standard errors. The change reported is the average marginal effect of the period,

$$\Delta = \frac{1}{T} \sum_{t=1}^{T} \Big[\hat{P}\big(y_t = 1 \mid \text{2023–2026}, x_t\big) - \hat{P}\big(y_t = 1 \mid \text{2014–2019}, x_t\big)\Big], \tag{11}$$

where x_t holds the actor and the language of text t; its 95% confidence interval is obtained by drawing the coefficients 400 times from their estimated distribution. This is the prevalence part of a structural topic model (Roberts et al., 2014), applied to the topics found by BERTopic.

5.2 Findings

The thematic map shows that the CBDC literature is driven by two themes: blockchain technology and the digital transformation of financial services, together with monetary policy and payments. Fintech and digital finance form a basic theme, while digital payments, India and the cashless society are niche topics.



1 (motor): digital transformation in financial services, monetary policy, monetary policy and economic impact, banking stability, regulation, efficiency

2 (motor): blockchain technology applications and security, digital transformation in law, digital ruble, cryptocurrency

3 (basic): fintech, crowdfunding, digital finance, blockchain, fintech, financial regulation

4 (emerging or declining): global financial crisis and policies, international payments, stablecoin, monetary sovereignty

5 (niche): financial inclusion, digital payments, india, cashless society

Figure 3. Thematic map of the keywords of research by economists

From our analysis, we observe a shift in the way economists discuss CBDC from 2014 to 2026. The early literature treated CBDC as a monetary policy instrument: half of the papers published before 2020 deal with monetary policy and its economic impact, against a fifth today. The topic model shows the same trend, as the topic on interest rates and uncertainty decreases by 9.6 points. Early papers also often discussed CBDC together with private cryptocurrencies: the largest topic of the period is characterized by words such as *cryptocurrencies* and *private*, and it is also the topic that lost the most ground since then (−13.2 points).

After 2020, as central banks started to work on their own projects, the discussion became more national. The digital euro and the digital ruble, almost absent before 2020, now appear in 11% and 15% of the papers, and *euro*, *ruble* and *India* are among the words that best characterize recent years. At the same time, financial inclusion is the theme that grows the fastest, from 6% to 16% of the papers. Overall, the literature moved from general questions on the effects of a CBDC to questions on specific national projects and on who they are meant for.

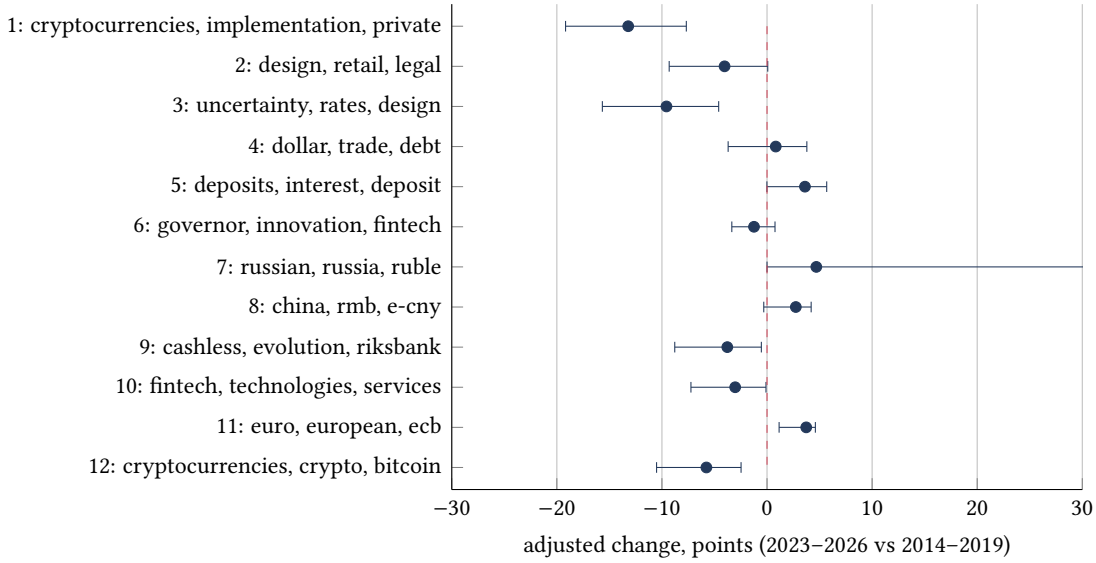


Figure 4. Change in the share of the twelve largest topics, adjusted for the type of author and the language, with 95% confidence intervals (axis cut at ±30 points: topics almost absent in 2014–2019 have very wide intervals)

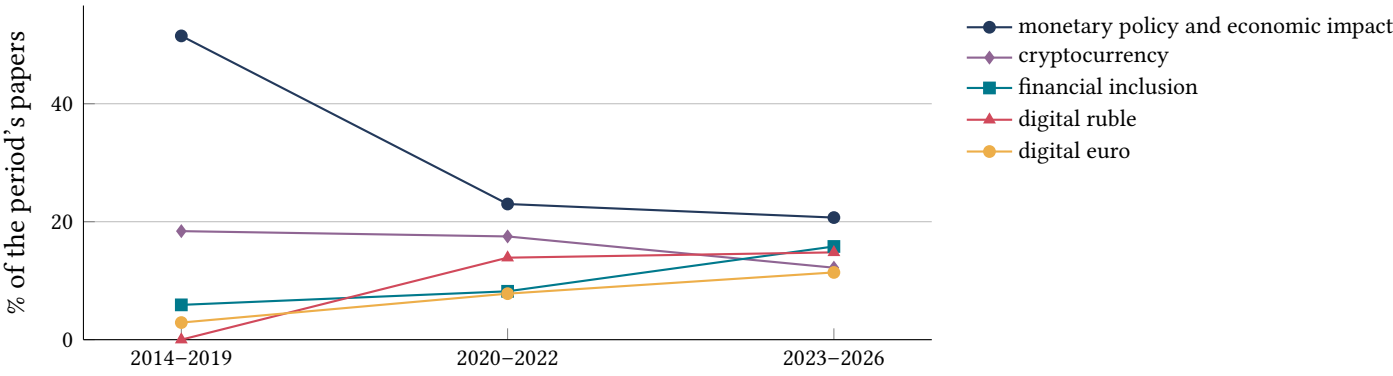


Figure 5. Share of research papers carrying each keyword, by period (136 papers in 2014–2019, 755 papers in 2020–2022, 2,919 papers in 2023–2026)

6 How economists discuss CBDCs: tone and stance

6.1 Method: tone (FinBERT) and stance (reading and a local language model)

6.2 Findings

6.3 Do policy events move the tone?

7 Why: stated motivations

7.1 Method: a validated codebook

7.2 Findings

8 When: the policy environment

8.1 Method: bursts, event study with placebo dates, difference-in-differences

8.2 Findings

9 The discussion and its environment

In this section, we put together the previous results period by period, and we discuss what they allow us to say, or not, about the causes of the changes we observe.

9.1 Period by period

From our analysis, we observe that between 2014 and 2019, economists wrote little about CBDCs, with 53 texts a year on average. The discussion was mainly about monetary policy: it is the most frequent motivation, and keywords such as monetary policy and economic theories suddenly become much more frequent during this period. The main event of this period, the announcement of Facebook's project Libra in June 2019, is not followed by any rise in attention. In addition, the tone of the texts remains cautious: the net tone is only 0.06 for research economists and 0.10 for central banks.

Between 2020 and 2022, the number of texts increases to 324 a year. This is the period corresponding to the first central bank reports and launches, but only two of the seven events of the period are followed by a significant rise in attention: the ECB report on a digital euro and the launch of the Sand Dollar in the Bahamas, both in October 2020. At the same time, the discussion becomes more favorable, especially among central banks, whose net tone rises to 0.22.

Finally, since 2023, economists write about 840 texts a year, and the discussion is now about national projects. Out of the four events of this period, only two lead to more texts: the start of the ECB preparation phase and, just barely, the EU bill on the digital euro. Keywords such as India, its Unified Payments Interface and monetary sovereignty also become much more frequent. This is also the period when texts are the most positive about CBDCs: compared with 2014-2019, the share of texts in favor of CBDCs rose by 40 points for central banks and by 14 points for research economists. In short, economists moved from a theoretical question on monetary policy to real national projects and became more positive as these projects moved forward.

9.2 What can and cannot be said about causes

Our analysis is purely descriptive: it allows us to see what changes and what moves at the same time, but not to prove what causes these changes. To see how far we can go, we test three links. The first one is between events and attention: 4 of the 12 policy events are followed by more texts than 95% of random dates. In other words, economists write more after some events, but this only tells us about

timing and not about causes. Then, by testing the link between events and tone, we observe that none of the 12 events changes the tone of the texts: the tone rises slowly over the whole period, without any sudden change after a specific event. Finally, we observe that after the sanctions on Russia in 2022, the share of texts about Russia, or written in Russian, that mention geopolitics and sanctions rose by 11.6 points more than in other texts. However, three other motivations also changed in these texts (financial inclusion, financial stability and monetary policy), so this result fits with an effect of the sanctions but does not prove it.

10 Robustness

11 Discussion

11.1 Limitations

12 Conclusion

A Search vocabulary

A single vocabulary drives both the searches and the relevance score (Section 3). It has 31 concepts, each written as a case-insensitive regular expression, and 115 search phrases derived from them, in 12 languages. Three principles guided it.

1. **Recall first.** The vocabulary covers the generic term in every major language, the names of national projects, the BIS Innovation Hub experiments, and the older names used before "CBDC" became standard (*digital fiat currency*, *central bank cryptocurrency*, *Fedcoin*, *RSCoin*, *CADcoin*, *digital base money*).

2. **Strong and ambiguous terms.** A *strong* term identifies a central bank digital currency on its own (*central bank digital currency*, *digital euro*, *e-CNY*). An *ambiguous* term does not: "CBDC" is also *chronic bullous dermatosis of childhood*, "DCEP" a chemical compound, "Project Stella" a satellite. Ambiguous terms count only when the document also contains monetary context: *currency, monetary, payment, central bank, banking, ledger, stablecoin*, and their equivalents in French, Spanish, Portuguese, German, Italian, Russian, Chinese, Japanese and Korean. When the only evidence is in the body of the text, a stricter context list is required (*central bank, digital currency, monetary, payments*).

3. **Never-scoring terms.** *Digital dollar* and *Sand Dollar* are reported but never scored: they are too often used for private tokens or unrelated products.

Table 3. Concepts of the search vocabulary (selection)

Concept	Forms matched
Generic	CBDC, wCBDC, rCBDC, mCBDC; central bank(s) digital currency/currencies; digital central bank money; tokenised central bank money or reserves
Other languages	monnaie numérique de banque centrale (MNBC); moneda / moeda digital de banco central; digitales Zentralbankgeld; valuta / moneta digitale di banca centrale; Japanese, Korean and Russian equivalents (*tsifrovaya valyuta tsentral'nogo banka*)
Euro area	digital euro, euro numérique, digitaler Euro, euro digitale, euro digital
China	e-CNY, digital yuan / renminbi, DC/EP (Digital Currency Electronic Payment), Chinese forms
Other projects	digital pound (Britcoin), e-krona, digital / e-rupee, eNaira, DCash, JAM-DEX, digital ruble (*tsifrovoy rubl'*), Drex / real digital, e-hryvnia, digital yen, e-HKD, e-peso, Bakong, e-Cedi, digital shekel, lira, won, rial, e-dirham
Early names	digital fiat currency, central bank cryptocurrency, central bank e-money / electronic money / digital cash, sovereign or national digital currency, digital legal tender, Fedcoin, RSCoin, CADcoin
BIS projects	mBridge; Projects Helvetia, Jura, Dunbar, Icebreaker, Aurum, Rosalind, Tourbillon, Mariana, Hamilton, Stella, Ubin, Inthanon, Khokha, Jasper, Agorá, Orchid, Polaris, Sela, Mandala, Ensemble
Adjacent	tokenised deposits, unified ledger (ambiguous: counted only with monetary context)

The full list of regular expressions and search phrases is in `cbdc_biblio/config.py`.

B Harvesting algorithms

B.1 Sources

Records come from four families of sources (Table 4). Nothing is filtered at harvest time: every record found by a search is kept, and relevance is decided afterwards on the merged corpus. Every step writes a checkpoint, so an interrupted harvest resumes where it stopped.

Academic databases. OpenAlex (journal articles, working papers, books, chapters, preprints, reports), Semantic Scholar, arXiv, HAL and DOAJ. Each search phrase is queried in titles and abstracts; results are merged on the source's identifier. OpenAlex reference lists are resolved into citation strings, which the co-citation analysis uses.

Central bankers' speeches. The BIS bulk archive (all central banks, 1996 onwards, full text), the ECB's speech dataset (full text, including non-English originals), and the BIS RSS feed for speeches published after the bulk archive.

Authorities. A crawl starting from 26 CBDC pages of central banks, EU institutions and international organisations, following links one level deep within the same site, PDFs included.

Press. Google News, searched through its RSS interface in 55 combinations of country edition, language and query. GDELT was also queried but throttled the connection for hours after a few requests, so it contributed nothing to the final corpus.

Table 4. Records by source and filtering stage

Source	Collected	CBDC a theme	By economists
Google News	24,309	9,496	0
OpenAlex	20,596	7,266	4,043
Semantic Scholar	12,528	4,000	2,164
DOAJ	1,750	485	251
BIS central bankers' speeches	618	465	465
Central banks and authorities (crawl)	266	239	205
HAL	152	83	31
ECB speeches	127	108	108
arXiv	93	90	24
Distinct records	48,188	17,709	4,710

A record found in several sources counts once per source; the last row counts distinct records.

B.2 Pagination and rate limits

Academic APIs return results page by page; the harvester follows the API's cursor (200 records per page for OpenAlex) until no page is left. Requests are spaced by a minimum interval Δ,

$$t_{k+1} - t_k \geq \Delta,$$

with $\Delta = 0.12$ s for OpenAlex, 3.5 s for arXiv and 1.2 s for Google News. After the j-th consecutive refusal (HTTP 429 or 5xx), the harvester waits $w_j = \min(60j, 900)$ seconds before retrying.

B.3 Press: recursive bisection of time windows

A Google News search returns at most about 100 items, whereas a single month can hold thousands of articles. Each search is therefore run on time windows that are split until every window falls below the cap:

1. Start from monthly windows $W = [a, b)$ covering January 2000 to the harvest date, on a stack.
2. Pop a window and fetch the items published in it.
3. If it returns $N(W) \geq 95$ items and lasts more than one day, split it at $m = a + (b - a)/2$, push $[m, b)$ and $[a, m)$, and go back to step 2.

4. Otherwise save the items and checkpoint the window.

A month is split at most $\lceil \log_2 31 \rceil = 5$ times, and keeping L windows costs $2L - 1$ requests. Google News limits requests over time, and the harvest was stopped with about 15 of the 55 searches unfinished; press counts are therefore incomplete, and skewed towards recent years.

B.4 Full text, enrichment and de-duplication

Full text. PDFs are extracted with PyMuPDF and web pages with trafilatura, which also recovers the page's date, author and outlet when the record lacks them. Texts are split into introduction, conclusion and section headings; for speeches and press, the opening and closing paragraphs play these roles.

Enrichment. Language is detected with `langdetect`; keywords are extracted with YAKE (one to three words) when a record has no author or indexed keywords. Dates are kept with their precision (day, month, year), and 1 January dates are flagged, since they often stand for a year only.

De-duplication. Records are matched on a key: the DOI for academic records, else their normalised title and year; the date and normalised title for speeches; the URL for other records. A second pass merges academic records with the same normalised title and year but different DOIs (a preprint and its published version, for instance). Merged records keep the most precise date, the longest abstract and the union of keywords and sources.

B.5 Relevance score

Each available field f of a record receives a centrality $c_f \in [0, 1]$: 1 for a title containing a strong term; 0.4 for a keyword list containing one (keywords are partly machine-assigned); for the abstract, the share of its sentences that contain one; for the body, half the density of mentions (per 1,000 words, saturating at 3) and half their spread (the share of ten equal slices of the text that contain one). The score is

$$\text{score} = 100\Big(0.6 \max_{f \in F} c_f + 0.4 \frac{1}{|F|} \sum_{f \in F} c_f\Big),$$

where F is the set of fields available, so a missing abstract or full text is not penalised. Records scoring at least 50 are labelled *main theme*, from 25 to 49 *secondary theme*, and below 25 *mentioned only*. Records whose only matches are ambiguous terms without monetary context are *ambiguous terms only*; records without any match are *off-topic*; press items whose text could not be retrieved are *unverified*. Records with CBDC as a theme but dated before 2014 were all homonyms or misdated, and are labelled *dated before 2014*. Table 5 gives the labels and their validation.

Table 5. Relevance labels and their validation

Label	Records	Read	Theme	95% CI
main theme	15,464	30	29	0.83–0.99
secondary theme	2,245	29	11	0.23–0.56
mentioned only	489	30	1	0.01–0.17
unverified (no text)	13,446	19	2	0.03–0.31
ambiguous terms only	1,709	10	3	0.11–0.60
off-topic	14,803	20	0	0.00–0.16
dated before 2014	32	0	0	–

Read: records of the stratified sample (138 texts) read blind to the label; Theme: read as having CBDC as a theme; Wilson intervals on the share.

B.6 Who counts as an economist

For each academic record with CBDC as a theme, the OpenAlex field of its primary topic and the institutions of its authors were retrieved (`scripts/tag_economists.py`). A text is attributed to economists when it is research classified in *Economics, Econometrics and Finance*, or research with an author at a central bank, the IMF or the BIS, or, for records absent from OpenAlex, research published in an economics venue; when it is a central bank speech; or when it is a document of a central bank or an international financial institution. Tables 6 and 7 show the result.

Table 6. Who counts as an economist

Texts with CBDC as a theme	Texts	
Research classified in Economics, Econometrics and Finance	3,783	kept
Research in another field, with a central bank, IMF or BIS author	148	kept
Research without OpenAlex record, in an economics venue	63	kept
Research in other fields	3,469	dropped
Central bank speeches	511	kept
Documents of central banks and international institutions	205	kept
Documents of governments and parliaments	34	dropped
Press articles	9,496	dropped
Texts by economists	4,710	

Field: OpenAlex primary topic of the work; author institutions from OpenAlex and from the record's affiliations.

Table 7. Research with CBDC as a theme left out by the economist filter, by field

OpenAlex field of research dropped	Texts
Computer Science	1,769
Business, Management and Accounting	663
no OpenAlex record	473
Social Sciences	281
Decision Sciences	244
Engineering	14
Medicine	7
Health Professions	5
Environmental Science	4
Arts and Humanities	3
Agricultural and Biological Sciences	2
Neuroscience	2
Immunology and Microbiology	1
Materials Science	1

C Validation samples and coding guide

Every measure used in the paper was compared with a reading of the texts. The samples were drawn at random; the motivation, stance and tone samples were topped up with texts written by economists so that each holds at least 50 of them (Table 8). Coding was done blind to the machine labels, by one coder assisted by a language model.

C.1 Coding guide

Relevance. Is a central bank digital currency the theme of the text, or one of its main themes? A passing mention, a homonym, or a text about cryptocurrencies or stablecoins only is coded *no*.

Table 8. Hand-coded validation samples

Sample	Units	From economists' texts
Relevance	140	25
Motivations, development	100	60
Motivations, test (held out)	74	50
Stance	177	100
Tone (sentences)	136	90

Motivations. Unit: the first 1,500 characters of the introduction, else of the abstract, else of the text. A motivation is coded when the text raises it as a reason to study, issue or worry about a CBDC; a text can have several. The ten motivations are defined as follows.

- *Decline of cash*: cash use is falling, or access to cash or public money is at risk.
- *Financial inclusion*: reaching unbanked or underbanked people; access to financial services.
- *Monetary sovereignty*: a threat from private money, stablecoins, Big Tech, cryptocurrencies or foreign currencies to the central bank's control of money.
- *Payment efficiency*: cheaper, faster, more efficient, resilient or innovative domestic payments.
- *Cross-border payments*: international payments, remittances, cross-border settlement.
- *Monetary policy*: monetary policy, interest rates, transmission, money supply.
- *Financial stability*: bank disintermediation, bank runs, deposit outflows, risks to financial stability.
- *Privacy and surveillance*: privacy, anonymity, data protection, surveillance or control of users.
- *Geopolitics and sanctions*: geopolitics, sanctions, dollar dominance, internationalisation of a currency.
- *Technology*: blockchain, distributed ledgers, tokenisation, smart contracts, programmable money.

Stance. Unit: the title and abstract (research), or the title and the first five sentences that name a CBDC (speeches, documents). *Favourable*: supports CBDCs or presents them positively; *critical*: opposes them or stresses their risks and problems; *neutral*: descriptive or balanced. Analysing risks is neutral; arguing that they can be solved so that a CBDC can be issued, or recommending issuance, is favourable.

Tone. Unit: one English sentence that names a CBDC. *Positive* or *negative* when the sentence presents CBDCs or their effects favourably or unfavourably; *neutral* when it describes, defines or reports.

C.2 Results

The motivation codebook was written on the 100 development texts only and evaluated on the held-out test texts (Table 9). Each alternative coder was tested on the same texts (Tables 10 and 11); thresholds and classifiers were set on the development texts (motivations) or by 5-fold cross-validation (stance). Agreement is measured by Cohen's $\kappa = (p_o - p_e)/(1 - p_e)$, where p_o is the observed and p_e the chance agreement. The language model's stance labels are used only for changes over time: it labels too many texts favourable, a bias that cancels in a difference if it is stable. Stance levels come from reading (Table 13).

Table 9. Motivation codebook on the held-out texts written by economists

Motivation	Read	Coded	Precision	Recall	κ
Decline of cash	7	1	1.00	0.14	0.22
Financial inclusion	7	3	1.00	0.43	0.56
Monetary sovereignty	6	1	1.00	0.17	0.26
Payment efficiency	16	3	0.67	0.12	0.12
Cross-border payments	1	3	0.33	1.00	0.48
Monetary policy	7	9	0.56	0.71	0.55
Financial stability	10	7	0.57	0.40	0.37
Privacy and surveillance	3	2	1.00	0.67	0.79
Geopolitics and sanctions	7	4	1.00	0.57	0.70
Technology (DLT, programmability)	11	11	0.73	0.73	0.65
All (pooled)	75	44	0.70	0.41	0.46

50 held-out texts by economists (of 74), never used to write the codebook. Read: texts where reading found the motivation; Coded: texts flagged by the codebook.

Table 10. Candidate coders for motivations

Coder	P	R	F_1	κ	κ, all
Codebook (regex)	0.70	0.41	0.52	0.46	0.50
Codebook OR classifier	0.42	0.60	0.50	0.39	0.42
SBERT, classifier (100 labels)	0.36	0.37	0.37	0.25	0.28
Local LLM, Qwen2.5 7B	0.30	0.61	0.40	0.25	0.28
Local LLM, Qwen2.5 3B	0.31	0.23	0.26	0.15	0.16
SBERT, definitions (zero-shot)	0.21	0.71	0.32	0.12	0.15
NLI zero-shot (mDeBERTa)	0.22	0.39	0.28	0.11	0.14

Scores on the 50 texts written by economists; last column: κ on all 74 hand-coded texts. Thresholds and classifiers set on the 100 development texts.

Table 11. Candidate coders for stance

Coder	Accuracy	κ	κ, all
Local LLM, Qwen2.5 7B	0.70	0.45	0.41
FinBERT tone	0.65	0.31	0.20
NLI zero-shot (mDeBERTa)	0.63	0.29	0.18
SBERT classifier	0.61	0.21	0.17

Scores on the 100 texts written by economists; last column: κ on all 177 hand-coded texts. Thresholds and classifiers by 5-fold cross-validation.

Table 12. FinBERT tone against reading

FinBERT ↓ / reading →	negative	neutral	positive	Precision
negative	17	11	2	0.57
neutral	2	25	3	0.83
positive	1	10	19	0.63

90 sentences from economists' texts, 30 per FinBERT label, read blind to the label: agreement 68%, $\kappa = 0.52$.

Table 13. Stance by reading, texts written by economists

Actor	n	Favourable	Neutral	Critical
Research economists	50	8 [3–19]	90 [79–96]	2 [0–10]
Central banks and institutions	50	52 [39–65]	42 [29–56]	6 [2–16]

Percentages with Wilson 95% intervals.

D Additional tables and figures

Table 14 lists the policy events used in the event study and the test of tone (Sections 6 and 8), with the attention each one received. Table 15 lists every topic found by the topic model (Section 5).

Table 14. Policy events and the attention they received

Date	Event	Texts	Expected	Ratio	p
18 Jun 2019	Facebook announces Libra	1	3.4	0.3	0.83
02 Oct 2020	ECB digital euro report	13	4.5	2.9	0.01
20 Oct 2020	Bahamas launches Sand Dollar	17	5.7	3.0	0.01
14 Jul 2021	ECB investigation phase	6	13.0	0.5	0.93
25 Oct 2021	Nigeria launches eNaira	9	11.4	0.8	0.69
20 Jan 2022	Fed CBDC discussion paper	14	10.1	1.4	0.24
09 Mar 2022	US order on digital assets	6	10.3	0.6	0.84
01 Nov 2022	India e-rupee pilot	14	14.6	1.0	0.55
07 Feb 2023	UK digital pound consultation	10	13.8	0.7	0.76
28 Jun 2023	EU digital euro bill	32	19.3	1.7	0.05
18 Oct 2023	ECB preparation phase	63	19.1	3.3	0.00
23 Jan 2025	US order banning a US CBDC	26	21.9	1.2	0.29

Texts by economists dated to the day, in the 15 days from the event; expected from the daily rate over days −90 to −15. p: share of 500 random dates with a larger $z = (N - E)/\sqrt{E}$.

Table 15. Topics of the economists' corpus

	Topic (top terms)	Texts	2014–2019	2020–2022	2023–2026
1	cryptocurrencies, implementation, private, traditional, national	470	18.5	13.5	8.5
2	design, retail, legal, private, privacy	439	10.4	11.2	8.6
3	uncertainty, rates, design, rate, interest	349	14.2	8.7	6.7
4	dollar, trade, debt, reserve, inflation	326	5.7	5.8	7.5
5	deposits, interest, deposit, rate, effects	319	2.8	5.0	7.3
6	governor, innovation, fintech, kong, hong	306	17.5	11.2	4.7
7	russian, russia, ruble, federation, legal	238	0.0	5.1	4.9
8	china, rmb, e-cny, yuan, internationalization	213	1.4	5.3	4.6
9	cashless, evolution, riksbank, forms, physical	204	6.6	4.9	4.1
10	fintech, technologies, services, innovation, finance	199	7.1	2.9	4.4
11	euro, european, ecb, area, legal	186	0.5	3.6	4.2
12	cryptocurrencies, crypto, bitcoin, cryptocurrency, assets	186	9.0	4.0	3.6
13	journal, list, approved, chapter, implementation	176	0.5	3.4	4.2
14	india, rupee, indian, rbi, upi	132	0.0	1.1	3.5
15	ruble, russia, russian, form, platform	130	0.0	1.9	3.2
16	cross-border, interoperability, settlement, instant, mbridge	127	0.9	2.1	2.9
17	euro, european, europe, ecb, parliament	127	0.9	2.4	3.0
18	euro, sie, european, auswahl, textabschnitt	110	0.5	1.3	2.8
19	nigeria, enaira, african, africa, inclusion	103	0.5	1.6	2.4
20	legal, assets, virtual, law, regulation	101	0.9	1.5	2.3
21	euro, project, rulebook, latest, in-depth information	77	0.5	0.6	2.1
22	raqamli, markaziy, circular, tahlil, joriy	73	0.0	0.7	1.6
23	ukraine, e-hryvnia, ukrainian, electronic, civil	69	1.4	1.3	1.6
24	ruble, legal, civil, agreement, platform	50	0.0	0.6	1.3

BERTopic on 4,710 texts (multilingual SBERT embeddings, UMAP, HDBSCAN with leaf selection and at least 40 texts per topic); 50% of texts were outliers, assigned to the nearest topic. Terms: class-based TF-IDF on English texts. Period columns: % of the period's texts.

E Data and code availability

E.1 Data

The corpus is a single file, `data/cbdc_corpus.csv`: one row per document and 49 columns, in UTF-8. Every record keeps its issuer, its date with the precision of that date, its original language, its keywords (author, indexed and extracted), its abstract, introduction and conclusion when available, its full text, its relevance label and theme score, and the sources it was found in. The harvest ran from 8 to 11 October 2026; it covers 48,188 documents, of which 17,709 have CBDC as a theme and 4,710 were written by economists (Section B). The build step can also export it in two formats read by bibliometric software: a Scopus-style CSV (for `bibliometrix` and VOSviewer) and RIS.

Caches hold the results that are slow to recompute: sentence embeddings, sentence tones, the language model's stance labels, and the OpenAlex fields and institutions of each paper. The hand-coded validation samples and every coder's predictions are in `presentation/` (Table 8).

Many full texts are under copyright. The data are therefore shared with the members of the project and on request, not published openly. The metadata, identifiers (DOI, OpenAlex and source identifiers, URLs) and the code that rebuilds the corpus from the public sources are not restricted.

E.2 Code

All code is in Python and kept in a version-controlled repository, available from the authors on request. One command per step rebuilds the corpus, and every step resumes where it stopped:

1. `run.py harvest`: query every source (Section B);
2. `run.py fulltext`: download and extract the full texts;
3. `run.py build`: merge, de-duplicate, enrich and score relevance;
4. `scripts/tag_economists.py`: retrieve the OpenAlex field and institutions of each paper;
5. `scripts/validate_coders.py`: score every candidate coder against the hand-coded samples;
6. `scripts/make_appendix.py`: write the tables of this appendix (`paper/tables/`).

The analyses (networks, topics, motivations, tone, stance, event studies, robustness) are in `cbdc_biblio/analysis.py` and `cbdc_biblio/nlp.py`.

E.3 Software and models

Python 3.10 or later; the packages and their versions are listed in `requirements.txt`. The models are open and run locally:

- sentence embeddings: `paraphrase-multilingual-MiniLM-L12-v2` (Sentence-Transformers);
- tone: FinBERT (`ProsusAI/finbert`);
- natural language inference (tested coder): `mDeBERTa-v3-base-xnli-multilingual-nli-2mil7`;
- stance: Qwen2.5 7B Instruct, 4-bit, run with `mlx-lm` (Apple Silicon); Qwen2.5 3B was tested.

Everything ran on a laptop (Apple M1, 8 GB of memory). Coding new texts with the language model requires Apple Silicon; every other step runs on any machine once the caches are present.

F Use of artificial intelligence and statement of originality

F.1 Use of artificial intelligence

Artificial intelligence was used in two distinct ways in this work.

As research instruments. Several language models are part of the method and are described with it: multilingual sentence embeddings (topics), FinBERT (tone), a natural language inference model and two local Qwen2.5 models (tested as coders; the 7B model codes stance). Each of them was evaluated against a reading of the texts before being used, and their scores are reported in Section C. They run locally; no text of the corpus was sent to an external service for coding.

As assistants. A general-purpose AI assistant helped to write and debug the code that collects, cleans and analyses the corpus, to produce tables and figures, and to draft parts of the technical description in the appendices. The hand coding of the validation samples was AI-assisted: texts were read and coded blind to the machine labels, with the help of the assistant. For this reason, the validation scores measure agreement with an AI-assisted reading.

The research question, the choice and interpretation of the methods, the reading of the results and the conclusions are the authors'. Every number reported in the paper is computed by the code from the data, and the authors checked the text, the code and the results; they take full responsibility for the content of the paper.

F.2 Statement of originality

The authors declare that this paper is their own work. It does not reproduce, in whole or in part, text written by others without quotation marks and a reference; every idea, result, data source and software taken from other works is cited. The corpus consists of documents published by their authors; it is used for analysis only, and no full text from it is reproduced in the paper. The text produced with the help of an AI assistant was read, checked and edited by the authors, and was not copied from any other source.